

Common Pediatric Infectious Diseases (Part Two): Common Clinical Syndromes

Pharyngitis · Acute otitis media · Acute bacterial sinusitis · Skin & soft tissue infection · UTI | For each: organisms · signs & symptoms · diagnosis · complications · management

Yellow = stressed by the doctor · **Blue** = other high-yield · **Red underline** = clue that points to the diagnosis

1. Pharyngitis

1.1 Definition & Causes

- Infection of the **pharynx and/or tonsils**
- Age < 4 y → mainly **viral**; > 4 y → mainly **bacterial**

Viruses: 80% (most common)

- Rhinovirus · coronavirus · adenovirus · enterovirus · influenza · parainfluenza · RSV · EBV · HSV
- Cough, rhinitis, conjunctivitis → think viral
- Exudate can also be viral → **EBV, adenovirus**

Bacteria: 20%

- **GABHS (15%)** → severe complications; **scarlet fever (sandpaper) rash**
- Group C & G strep · *C. diphtheriae* · *Arcanobacterium haemolyticum* · *Chlamydophila pneumoniae* · *Mycoplasma pneumoniae* · *Yersinia enterocolitica* · *Francisella tularensis*

1.2 Modified Centor (Mclsaac) Criteria

Criterion	Points
Temperature > 38 °C	+1
Absence of cough	+1
Swollen, tender anterior cervical nodes (unilateral)	+1
Tonsillar swelling or exudate	+1
Age 3–14 y / 15–44 y / ≥ 45 y	+1 / 0 / -1

Score	≤ 0	1	2	3	≥ 4
Strep risk	1–2.5%	5–10%	11–17%	28–35%	51–53%

- **≥ 4 → mostly bacterial** · winter season → ↑ suspicion

1.3 Clinical Features: Viral vs Bacterial (GAS)

Feature	Viral	Bacterial (GAS)
Age	All ages	Peak 5–11 y
Onset	Variable	Sudden , severe fever
Sore throat	+	+++
Fever	Variable	+++
Abdominal pain	±	+ / ++
Cough, rhinitis, hoarseness, conjunctivitis, diarrhea	+	- (absent)
Other symptoms	Myalgia, arthralgia	Headache
Tonsillar hypertrophy	Variable	+++
Exudate	±	+++
Palatal petechiae	±	+
Cervical nodes	Bilateral, nontender	Unilateral, tender
Rash	Characteristic exanthems	Scarlet fever rash

1.4 Diagnostic Tests

Test	Sensitivity	Specificity	Note
Rapid antigen (RADT)	70–97% (↓)	> 95%	(+) → treat; (-) → culture
Throat culture	97%	99%	↑ sens + ↑ spec, but 2–3 days
Molecular (PCR)	accurate		expensive → not everywhere

- ↑ Specificity → rules **in**; ↑ sensitivity → rules **out** (screening)
- RADT is poor for screening → **negative RADT → throat culture**
- Culture delay is acceptable: **9-day window** to prevent rheumatic fever

1. Pharyngitis (continued)

1.5 Should We Treat GAS? & Clinical Approach

- Yes → **to prevent acute rheumatic fever** (9-day window)
- **Antibiotics do NOT prevent glomerulonephritis**
- Child looks well → RADT: (+) → start Tx; (-) → wait for culture
- No RADT/culture available → decide clinically (epidemiology + manifestations)

Clinical & epidemiological features

- ▶ Suggest **viral** → symptomatic therapy (no test needed)
- ▶ Suggest **bacterial** → test:
 - RADT (+) → antibiotics · RADT (-) → throat culture
 - Throat culture (+) → antibiotics · (-) → symptomatic
 - Molecular test (PCR) (+) → antibiotics · (-) → symptomatic

1.6 Complications of GAS Pharyngitis

Suppurative	Non-suppurative
Retropharyngeal abscess · peritonsillar abscess · cervical adenitis · otitis media · sinusitis · mastoiditis · bacteremia → metastatic infection	Acute rheumatic fever · acute glomerulonephritis · reactive arthritis (?) · toxin-mediated → streptococcal toxic shock syndrome

1.7 Treatment

- **Penicillin V** → GAS highly susceptible
- Alternative: **amoxicillin** → 10 days
- β -lactam allergy: **macrolide or clindamycin** → in severe/disseminated disease, confirm sensitivity first

1.8 Lecture Cases

Case 1: 3 y, low-grade fever, runny nose, cough, sore throat

- Age ≤ 3 y + cough + rhinitis → **viral** → **supportive**
- Tonsillar swelling + headache + unilateral tender cervical LN → would point to bacterial



Case 1: red pharynx, no exudate

Case 2: 6 y, high fever, severe sore throat, no cough

- Age > 3 y + **exudate + no cough** → **GAS** ($\pm$ scarlet fever)
- → antibiotic (**amoxicillin 10 d**) to prevent rheumatic fever



Case 2: tonsillar exudate

2. Acute Otitis Media (AOM)

2.1 Etiology

- **Bacteria 60–80%** → why we treat
 - ▶ **S. pneumoniae 30% = #1** despite vaccine (vaccine covers only invasive subtypes)
 - ▶ **H. influenzae 20%** → often amoxicillin-resistant → amox-clav; now ↓ by vaccination
 - ▶ **M. catarrhalis 20%**
- Viruses 20–40%
- **Spontaneous resolution 50%** (> 2 y)
- **Purulent conjunctivitis** → think **H. influenzae**

2.2 Pathophysiology

Viral URTI → secretions + edema → **Eustachian tube (ET) dysfunction = hallmark**

- ▶ ET blocked → **–ve middle-ear pressure** → reflux of nasopharyngeal microbes into ME
 - ▶ ↓ mucociliary defense · ↑ bacterial colonization & adhesion · altered immunity
 - ▶ Nose blowing / sniffing → pushes microbes into ME
 - ▶ Mucus accumulates in ME → medium for bacterial growth
 - ▶ Adenoids → bacterial reservoir?
- Ascending bacterial (± viral) infection via ET → **AOM**

2.3 Clinical Presentation

Symptoms

- Ear pain (**ear tugging** in infants)
- Irritability / crying
- Fever
- Ear discharge (± perforation)
- Diarrhea ± vomiting
- Bulging → only in acute suppurative OM; redness alone → not specific

Signs

- **Bulging TM** → **hallmark**
- Dull TM
- Absent light reflex
- ↓ / absent mobility
- Air-fluid level
- Otorrhea

2.4 Who to Treat & For How Long

Group	Action	Duration
< 2 y (esp. < 6 mo)	Always treat (↑ complications)	10 d (prevents failure)
≥ 2 y, severe : toxic, T ≥ 39 °C, severe otalgia, bilateral OM	Treat now	5 d (= 7 d)
≥ 2 y, mild	Supportive → reassess at 48–72 h : resolved → no antibiotic; persists → treat	5 d

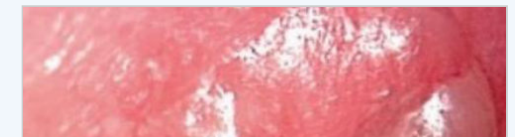
- Symptoms should improve by day 5

2.5 Antibiotic Choice (Flowchart)

- Amoxicillin in last 30 d **or** purulent conjunctivitis? (H. influenzae resistance)
- ▶ No → **HIGH-dose amoxicillin** → improved → 7 d (uncomplicated, > 2 y) / 10 d (others)
not improved → high-dose amox-clav
 - ▶ Yes → **high-dose amox-clav** → improved → 10 d
not improved → **IV ceftriaxone 50 mg/d × 3 d**
still no improvement → **tympanocentesis** → adjust per C/S (mainly immunocompromised)

2.6 Case 3

- 6-mo infant: fever, vomiting, excessive crying
- Ear exam: **bulging TM** → AOM
- < 2 y → **antibiotics now, 10 d**



Bulging TM

2.7 Complications

- TM perforation · conductive hearing loss · mastoiditis · labyrinthitis
- Brain abscess · epidural abscess
- **Otic hydrocephalus** → ↑ ICP without hydrocephalus

3. Acute Bacterial Sinusitis (ABS)

3.1 Pathogenesis

- ABS → **100% preceded by URTI** (AOM → ~90%)
- Sinus mucosa is continuous with nasal mucosa
- **< 4 y → no sinusitis** (sinuses underdeveloped)

3 keys to normal sinus physiology

1. **Patent ostia** → URTI secretions block the **ostiomeatal complex**
 2. **Ciliary function** → moves secretions to the nose; impaired (congenital, CF, acute URTI) → bacteria multiply
 3. **Secretion quality** → overproduction in URTI
- → Fluid accumulation → medium for bacteria → sinusitis
 - High risk: damaged cilia (e.g. **cystic fibrosis**)

3.2 Clinical Manifestations (3 Types)

Persistent	Nasal discharge and/or cough > 10 d, not improving
Severe	T ≥ 39 °C + purulent nasal discharge together > 3 d
Worsening	URTIs improving → worse on day 6–7 : new/recurrent fever or ↑ nasal symptoms/cough

3.3 Etiology

- **S. pneumoniae 30–40%** · H. influenzae 20% · M. catarrhalis 20% · viruses 10%
- Staphylococci, anaerobes, fungi → uncommon; seen in **immunocompromised**

3.4 Diagnosis

- **Clinical** is enough
- Uncomplicated → **no imaging** (only finding = mucosal thickening, also seen in viral URTI)
- **Sinus aspiration** → special cases only: not responding to Tx, immunocompromised

3.5 Who to Treat & Management (Flowchart)

Treat now if any:

- ▶ Severe or worsening symptoms
- ▶ Concurrent bacterial infection (AOM, pneumonia)
- ▶ Suspected complications
- ▶ Underlying disease: asthma, CF, ciliary dysfunction, immunodeficiency

None → outpatient observation **3 d** → improving: symptomatic · not improving: treat

Choosing the antibiotic

- ▶ Daycare, severe, or amoxicillin in last 30 d → **high-dose amox-clav**
- ▶ None of these → **high-dose amoxicillin** → not improving → amox-clav
- ▶ Improving → continue until **7 d after symptoms resolve** (total 7–10 d)
- ▶ Not improving on amox-clav → **sinus imaging / aspiration** (fungal in immunocompromised) → adjust per C/S

3.6 Complication

- **Brain abscess**

4. Skin & Soft Tissue Infections (SSTI) → mainly *S. aureus* + group A strep

4.1 Case 4 & Impetigo

- **Case 4:** 6 y, red swollen chin + **golden crusts** → impetigo
 - ▶ **Well & afebrile** → **topical** antibiotic (mupirocin)
 - ▶ **Ill & febrile** → **systemic** (narrow-spectrum, e.g. flucloxacillin)



Non-bullous: honey crusts



Bullous (*S. aureus*)

- Localized, **highly contagious**; age **2–5 y**
- **Non-bullous** → **GAS** → **golden/honey crusts**
- **Bullous** → ***S. aureus* toxin** (not the organism) → pustules rupture → erosion with collarette of scale → heals without scar

4.2 Folliculitis

- Hair follicle, **epidermis only** → more superficial & widespread than furuncle
- Discrete **2–5 mm papules/pustules** on red base; single or grouped; **hair shaft in center**
- ***S. aureus*** (predominant) ± GAS
- Often resolves spontaneously, no scar
- Mild → chlorhexidine cleanser, topical mupirocin/clindamycin, benzoyl peroxide

4.3 Furuncle & Carbuncle

- **Furuncle (boil):** hair follicle infection extending **deep into dermis**
- **Carbuncle:** painful aggregate of **contiguous follicles, multiple drainage points**, surrounding inflammation
- Almost always ***S. aureus***
- Tx: frequent **hot, moist compress** → drainage; large furuncles & most carbuncles → **surgical drainage**

4.4 Ecthyma & Ecthyma Gangrenosum

- **Ecthyma:** rare, **deep ulcer** into dermis → ***S. pyogenes*** → systemic anti-streptococcal antibiotic
- **Ecthyma gangrenosum:** **necrotic center**, serious, toxin-mediated → gram –ve, esp. ***Pseudomonas aeruginosa***
 - ▶ Immunocompromised / neutropenic (e.g. leukemia) → septicemia, can be fatal
 - ▶ → **Prompt anti-pseudomonal + blood culture**

4.5 Erysipelas vs Cellulitis

	Erysipelas	Cellulitis
Depth	Upper dermis + lymphatics (superficial)	Dermis + subcutaneous (deep)
Look	Sharply demarcated , follows lymphatic chain	Edema, warmth, erythema, tenderness
Organism	GAS (> staph); occasionally group B, C, G	<i>S. pyogenes</i> + <i>S. aureus</i>
Tx	Oral anti-strep, 7–14 d	Anti-strep + anti-staph (oral 5–7 d)

- Erythrasma is shown in the lecture photos with no further detail

5. Urinary Tract Infection (UTI)

5.1 Definition & Case 5

- **UTI = significant bacteriuria + symptoms**
- Asymptomatic bacteriuria → don't treat (except pregnancy / before procedures)
- **Infants: fever with no focus** is often the only sign
- **Case 5:** 9 mo, fever 3 d, well, T 38.9 °C, no focus → **urine analysis**
- Suspect UTI: fever ≥ 38 °C + no source, esp. with **age < 12 mo, female, uncircumcised boy, T > 39 °C, fever > 48 h**

5.2 Pathogenesis

- GI bacteria colonize periurethral mucosa → ascend → bladder → ureters → kidneys
- Bacterial virulence → e.g. Klebsiella adheres to ureter → kidney
- Host: obstruction + urostatics (stones, tumor, **constipation**), **neurogenic bladder**, VUR
- **Circumcision ↓ UTI ~8x** in boys < 3 mo

5.3 Etiology

- **E. coli 70–80%** (from anal area) · Klebsiella · Proteus · Pseudomonas · Enterococcus (gram +) · Enterobacter
- **GBS** → **neonates**
- **S. saprophyticus** → **adolescent girls** (gram +); S. aureus rare
- **Candida** → urinary catheter → remove catheter
- **Adenovirus** → **hemorrhagic cystitis** (immunocompromised)
- **TB** → **sterile pyuria** (↑ WBC/pus, culture –ve)
- **Saudi data:** E. coli + K. pneumoniae most common; resistance ampicillin ~70%, co-trimoxazole ~60% → **3rd-gen cephalosporin / aminoglycoside**

5.4 Clinical Features

Age	Most common	Less common	Least common
< 3 mo	Fever, vomiting, lethargy, irritability	Poor feeding, FTT	Abdominal pain, jaundice, hematuria, offensive urine
≥ 3 mo, preverbal	Fever	Abdominal pain, loin tenderness, vomiting, poor feeding	Lethargy, irritability, hematuria, offensive urine, FTT
≥ 3 mo, verbal	Frequency, dysuria	Dysfunctional voiding, continence change, abdominal pain, loin tenderness	Fever, malaise, vomiting, hematuria, offensive/cloudy urine

Lower UTI (cystitis) → bladder	Upper UTI (pyelonephritis) → kidney
Dysuria, urgency, frequency, suprapubic pain, hematuria	Fever , shivering, nausea/vomiting, diarrhea, flank pain + lower UTI symptoms
Tx 3–7 d	Tx 7–14 d

5.5 Urinalysis & Microscopy

- **Pyuria:** ≥ 5 WBC/HPF (centrifuged) or ≥ 10 WBC/mm³ (uncentrifuged)
- Urinalysis = **screening tool** (↑ sensitive) → all negative → UTI unlikely

Test	Sensitivity %	Specificity %
Leukocyte esterase (WBC breakdown)	83	78
Nitrite	53	98
LE or nitrite positive	93	72
Microscopy: WBCs	73	81
Microscopy: bacteria	81	83

5. Urinary Tract Infection (continued)

5.6 Urine Culture (Gold Standard) & Collection

Method	Use	Significant growth
Transurethral catheter (in/out)	1st choice: infants, not toilet-trained	> 50,000 CFU/mL (or > 10,000 with fever + pyuria)
Clean-catch midstream	1st choice: toilet-trained; 2nd choice if catheter impossible (e.g. bleeding tendency)	> 100,000 CFU/mL (or > 10,000 with fever + pyuria)
Suprapubic aspiration (from bladder)	Neonates, neurological pts; phimosis, severe labial adhesions	Any CFU (even one)

- **Urine bag X** → very high contamination → never for culture diagnosis; a **negative** result can help exclude UTI
- Colonization → from inside the body; contamination → from outside

5.7 Diagnostic Approach (Flowchart)

Suspect UTI: infant with fever + no focus, or older child with urinary symptoms
 → Collect by recommended method → **dipstick + microscopy + culture**

- ▶ **Febrile with urinary symptoms / unwell** → start empiric antibiotic now → adjust per culture
- ▶ **Afebrile with urinary symptoms / well with fever, no source** → check dipstick + microscopy:
 - LE, nitrite, pyuria or bacteriuria (+) → start empiric antibiotic, await culture
 - All (-) → wait for culture; repeat after 24–48 h if fever persists

5.8 Treatment

- **Oral = IV** in children ≥ 2 mo
- **Toxic** or can't take oral → IV until improved for 24–48 h → switch to oral
- Empiric → then **step down per culture**; check previous cultures

≥ 3 mo – 14 y	First line	Alternative*	Duration
Outpatient (oral)	1) Amox-clav 2) Cephalexin, cefuroxime, cefprozil	Cefixime (regular dose)	3–7 d (up to 10 d by severity/age)
Inpatient (IV): toxic, vomiting, no oral, pyelonephritis	Ceftriaxone	Gentamicin ± ceftriaxone	7–14 d

- *Alternative → no improvement 48 h after first choice, or first choice unsuitable
- **Gentamicin** → previous ESBL UTI or cephalosporin in last 3 mo
- **Ceftriaxone + aminoglycoside** → critically ill or worsening on first line

5.9 Imaging

First febrile UTI, < 3 y → **RBUS 2–6 wk** after infection (look for anomalies) → abnormal → **VCUG**

Any age + recurrent / complicated / febrile non-E. coli UTI / family history of VUR → **RBUS + VCUG** → abnormal → refer urology/nephrology

- Why wait 2–6 wk? Acute inflammation / debris → **false-positive US**
- **Early US** → severely ill, complicated, obstruction, or not improving by 48 h → detects renal/perirenal abscess, pyonephrosis
- **Recurrent** = ≥ 2 symptomatic UTIs in 12 mo
- **Complicated** = critically ill (urosepsis/shock), obstruction, or no response after 48 h of antibiotics
- **Refer** → recurrent UTI + abnormal imaging, VUR grade III–IV, congenital renal anomaly, bladder dysfunction, ↓ kidney function, hypertension

Summary

Syndrome	Main organisms	Hallmark	Diagnosis	Treatment	Complications
Pharyngitis	Viral 80%; GAS 15%	GAS: sudden high fever, exudate, unilateral tender nodes, no cough, scarlet rash; 5–11 y	Centor → RADT → (–) culture	Penicillin V / amoxicillin 10 d; allergy → macrolide/clindamycin	ARF (preventable), AGN, abscesses, STSS
AOM	<i>S. pneumoniae</i> 30%, <i>H. influenzae</i> 20%, <i>M. catarrhalis</i> 20%	Bulging TM ; ET dysfunction	Clinical (ear exam)	< 2 y: treat 10 d; ≥ 2 y: severe → 5 d, mild → wait 48–72 h. High-dose amoxicillin → amox-clav → IV ceftriaxone	Perforation, hearing loss, mastoiditis, abscess, otic hydrocephalus
ABS	<i>S. pneumoniae</i> 30–40%, <i>H. influenzae</i> 20%, <i>M. catarrhalis</i> 20%	100% after URTI; > 10 d / ≥ 39 °C + pus > 3 d / worse day 6–7	Clinical, no imaging	High-dose amoxicillin or amox-clav, 7 d after resolution (7–10 d)	Brain abscess
SSTI	<i>S. aureus</i> , GAS; <i>Pseudomonas</i> (ecthyma gangrenosum)	Honey crusts (impetigo); necrotic center (EG)	Clinical; blood culture in EG	Topical mupirocin → systemic flucloxacillin; drain boils	EG → septicemia
UTI	<i>E. coli</i> 70–80%, <i>Klebsiella</i>	Infant: fever, no focus; older: dysuria, frequency	UA screen + culture (gold); catheter > 50k, midstream > 100k, SPA any	Amox-clav oral 3–7 d; ceftriaxone IV 7–14 d	Renal abscess, pyonephrosis

Skin Lesions by Depth

Lesion	Depth	Organism	Clue	Treatment
Impetigo	Superficial	GAS (non-bullous), <i>S. aureus</i> toxin (bullous)	Honey crusts, 2–5 y, contagious	Topical → systemic if febrile
Folliculitis	Epidermis	<i>S. aureus</i>	Pustule with central hair shaft	Self-limited; topical
Furuncle	Deep dermis	<i>S. aureus</i>	Single boil	Hot compress ± drainage
Carbuncle	Dermis + surrounding tissue	<i>S. aureus</i>	Many follicles, multiple drainage points	Surgical drainage
Ecthyma	Ulcer into dermis	<i>S. pyogenes</i>	Deep ulcer	Systemic anti-strep
Ecthyma gangrenosum	Necrotic	<i>P. aeruginosa</i>	Necrotic center, neutropenia	Anti-pseudomonal + blood culture
Erysipelas	Upper dermis + lymphatics	GAS	Sharply demarcated	Oral anti-strep
Cellulitis	Dermis + subcutaneous	<i>S. pyogenes</i> , <i>S. aureus</i>	Warm, red, swollen, tender	Anti-strep + anti-staph

Ultra-Quick Differentiation (Keywords)

Keyword combination	Think of
≤ 3 y, cough, runny nose, conjunctivitis, sore throat	Viral pharyngitis → supportive
> 3–4 y, sudden high fever, no cough, exudate, unilateral tender nodes	GAS → penicillin V / amoxicillin
Sore throat + sandpaper rash	Scarlet fever (GAS)
Exudative pharyngitis, not strep	EBV, adenovirus
Centor ≥ 4	51–53% strep → mostly bacterial
Ear tugging, crying, fever + bulging TM	AOM
AOM + purulent conjunctivitis / amoxicillin in last 30 d	H. influenzae → amox-clav
↑ ICP without hydrocephalus after AOM	Otic hydrocephalus
Nasal discharge/cough > 10 d, not improving	Persistent ABS
≥ 39 °C + purulent discharge > 3 d	Severe ABS
URTI improving → new fever day 6–7	Worsening ABS
Recurrent sinus/ear infections, damaged cilia	Cystic fibrosis / ciliary dysfunction
Child < 4 y with "sinusitis"	Unlikely: sinuses underdeveloped

Keyword combination	Think of
Golden/honey crusts on the chin, 2–5 y	Non-bullous impetigo (GAS)
Bullae in an infant, heals with no scar	Bullous impetigo (S. aureus toxin)
Pustule with hair shaft in center	Folliculitis
Many draining follicles, painful	Carbuncle
Necrotic black center, neutropenic child	Ecthyma gangrenosum (Pseudomonas)
Superficial, sharply demarcated, along lymphatics	Erysipelas (GAS)
Well infant, fever, no focus	UTI → urine analysis
Adolescent girl, gram-positive UTI	S. saprophyticus
Hemorrhagic cystitis, immunocompromised	Adenovirus
Pyuria, sterile culture	TB
UTI in a neonate	GBS
Candida in urine + catheter	Remove the catheter
Nitrite (+) / LE or nitrite (+)	Best specificity 98% / best sensitivity 93%

Case-Based Practice MCQs (1–5 = lecture cases; answers on the next page)

1. 3-year-old: low-grade fever, runny nose, cough, sore throat. Best management?
 - A. Penicillin V for 10 days
 - B. Supportive treatment
 - C. RADT, amoxicillin if positive
 - D. Clindamycin
2. 6-year-old: high fever, severe sore throat, no cough, tonsillar exudate. Main reason to give antibiotics?
 - A. Shorten the illness
 - B. Prevent acute rheumatic fever
 - C. Prevent sinusitis only
 - D. Treat EBV
3. 6-month-old: fever, vomiting, excessive crying; ear exam shows a bulging TM. Best management?
 - A. Observe 48–72 h
 - B. Antibiotics now, 10 days
 - C. Antibiotics, 5 days
 - D. Tympanocentesis first
4. 6-year-old: red swollen chin with golden crusts; well and afebrile. Best treatment?
 - A. Topical mupirocin
 - B. IV flucloxacillin
 - C. Surgical drainage
 - D. Anti-pseudomonal antibiotic
5. 9-month-old: fever 3 days, looks well, T 38.9 °C, no focus. Next step?
 - A. Chest X-ray
 - B. Urine analysis
 - C. Throat culture
 - D. Reassure and discharge
6. Rapid antigen test for GAS: which is correct?
 - A. Sensitivity 97%, so (–) excludes GAS
 - B. Specificity > 95%, lower sensitivity → (–) needs culture
 - C. Takes 2–3 days
 - D. More accurate than culture
7. 3-year-old with AOM took amoxicillin 3 weeks ago. First-line now?
 - A. High-dose amoxicillin
 - B. High-dose amoxicillin-clavulanate
 - C. IV ceftriaxone
 - D. Azithromycin
8. Hallmark sign of AOM?
 - A. Red TM
 - B. Bulging TM
 - C. Ear tugging
 - D. Fever
9. Nasal discharge and cough for 12 days, not improving, afebrile. Diagnosis?
 - A. Viral URTI, reassure
 - B. Persistent acute bacterial sinusitis
 - C. Severe acute bacterial sinusitis
 - D. Needs sinus CT
10. Neutropenic child with leukemia: skin lesion with necrotic black center. Organism?
 - A. *S. aureus*
 - B. *S. pyogenes*
 - C. *Pseudomonas aeruginosa*
 - D. Group B strep
11. 10-month-old, not toilet-trained. Collection method and cut-off?
 - A. Urine bag, any growth
 - B. Transurethral catheter, > 50,000 CFU/mL
 - C. Midstream, > 100,000 CFU/mL
 - D. Suprapubic, > 100,000 CFU/mL
12. 4-year-old with febrile UTI, well, drinking. Correct statement?
 - A. IV beats oral
 - B. Oral = IV in children ≥ 2 months
 - C. Ampicillin preferred in Saudi children
 - D. Co-trimoxazole is first line
13. 1-year-old recovered from first febrile UTI. When is RBUS done?
 - A. Immediately, during fever
 - B. 2–6 weeks after infection
 - C. Only after a 2nd UTI
 - D. Never under 3 years
14. Pyuria with repeatedly sterile cultures. Consider?
 - A. *E. coli*
 - B. Adenovirus
 - C. Tuberculosis
 - D. Candida

Answer Key

1	B: ≤ 3 y + cough + rhinitis $\rightarrow$ viral $\rightarrow$ supportive
2	B: prevent ARF (9-day window); GN not preventable
3	B: < 2 y (esp. < 6 mo) $\rightarrow$ treat, 10 d
4	A: well/afebrile $\rightarrow$ topical; ill/febrile $\rightarrow$ systemic
5	B: infant fever without focus $\rightarrow$ UTI $\rightarrow$ urine analysis
6	B: RADT sens 70–97%, spec $> 95\%$ $\rightarrow$ (–) $\rightarrow$ culture
7	B: amoxicillin in last 30 d $\rightarrow$ amox-clav (H. influenzae)
8	B: bulging TM; redness is non-specific
9	B: > 10 d, not improving $\rightarrow$ persistent; no imaging
10	C: ecthyma gangrenosum $\rightarrow$ anti-pseudomonal + blood culture
11	B: catheter $> 50,000$ (or $> 10,000$ + fever + pyuria)
12	B: oral = IV ≥ 2 mo; ampicillin $\sim 70\%$ and co-trimoxazole $\sim 60\%$ resistant
13	B: 2–6 wk avoids false positives; early only if severe/complicated
14	C: TB $\rightarrow$ sterile pyuria